SOFTWARE ENGINEERING ARCHITECTURE DOCUMENT

Complete Data Flow Diagram (DFD) Specification: Level 0, Level 1 & Level 2

Full Multi-Level Functional Decomposition, Data Stores, Data Dictionaries & Cloud Flow

Project Title:	CareerForge AI — Autonomous Career Readiness Platform
DFD Hierarchy:	• **Level 0:** Context Level Diagram (System Boundaries & External Entities) • **Level 1:** Top-Level Functional Decomposition (Processes 1.0 to 6.0 & Stores D1-D4) • **Level 2:** Deep Functional Decomposition of AI Interview Engine (Process 4.0)
External Entities:	1. Student (Candidate), 2. Faculty / Admin (TPO / HOD), 3. Google Gemini 3.6 Flash AI, 4. Supabase Cloud (PostgreSQL)
Data Stores:	D1: auth.users, D2: public.student_states, D3: domain_catalog, D4: cert_registry
Standard Followed:	Yourdon & Coad / Gane & Sarson Software Engineering Notation

Abstract: A Data Flow Diagram (DFD) visualizes how data enters, flows through, is stored, and is transformed by an application. This document provides the complete, authoritative multi-level DFD for CareerForge AI, detailing every input stream, transformation process, datastore mutation, and output deliverable across both the Student and Faculty Admin domains.

1. DFD Level 0 — Context Level Diagram

The Level 0 Context Diagram establishes the **system boundary** of CareerForge AI. The entire software platform is treated as a single high-level process (0.0 CareerForge AI Platform) interacting with four external entities:

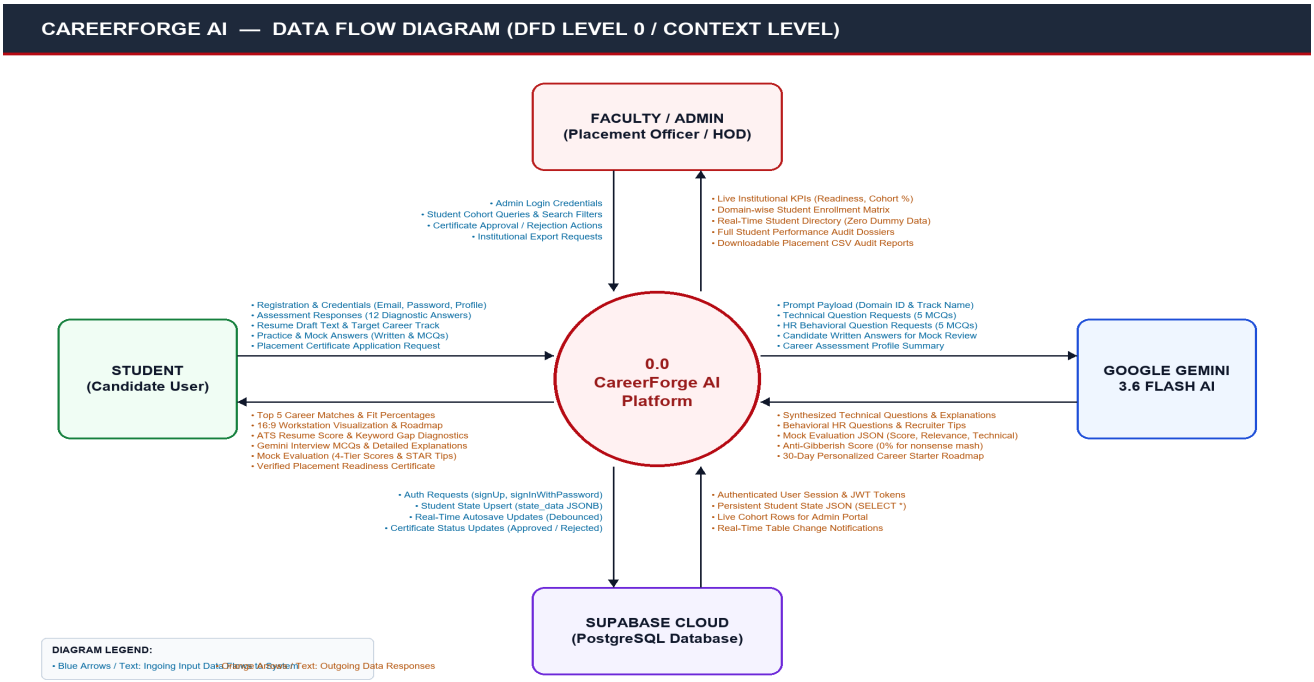


Figure 1.1: CareerForge AI — Level 0 Context Data Flow Diagram

Level 0 Entity & Data Flow Table:

Entity	Role in System	Inputs from Entity to System	Outputs from System to Entity
Student	Primary Candidate	• Credentials (Email, Password) • Profile (Name, Roll, Dept, Year) • 12 Assessment Answers • Resume Draft & Target Track • Practice & Mock Answers • Certificate Application Request	• Top 5 Career Matches & Fit % • 16:9 Workstation Visual & Roadmap • ATS Score & Missing Keywords • Gemini MCQs & Explanations • Mock 4-Tier Scores & STAR Tips • Verified Digital Certificate
Faculty / Admin	Academic Oversight	• Admin Login Credentials • Cohort Search & Multi-Filters • Certificate Approvals/Rejections • Institutional Export Requests	• Live Institutional KPIs (Readiness %) • Domain Enrollment Matrix • Real-Time Student Directory Table • Student Performance Dossiers • Accreditation CSV Report Download
Google Gemini 3.6 Flash AI	External LLM Engine	• Question Generation Prompts (Domain) • Candidate Written Mock Answers • Assessment Profile Summary	• 5 Synthesized Technical MCQs • 5 Behavioral HR MCQs & Tips • Mock 4-Tier Scoring JSON • Anti-Gibberish Flag (0% Score) • 30-Day Starter Career Roadmap
Supabase PostgreSQL	Cloud BaaS & DB	• Auth Requests (signUp, signIn) • State Upsert (JSONB state_data) • Real-time Autosave Sync • Certificate Endorsement Updates	• Authenticated JWT Session Token • Restored Student State (SELECT *) • Live Cohort Rows for Admin • Real-Time Database Change Stream

2. DFD Level 1 — Functional Decomposition

Level 1 decomposes the central system into **six core sub-processes** and identifies the **four primary data stores** governing the application lifecycle:

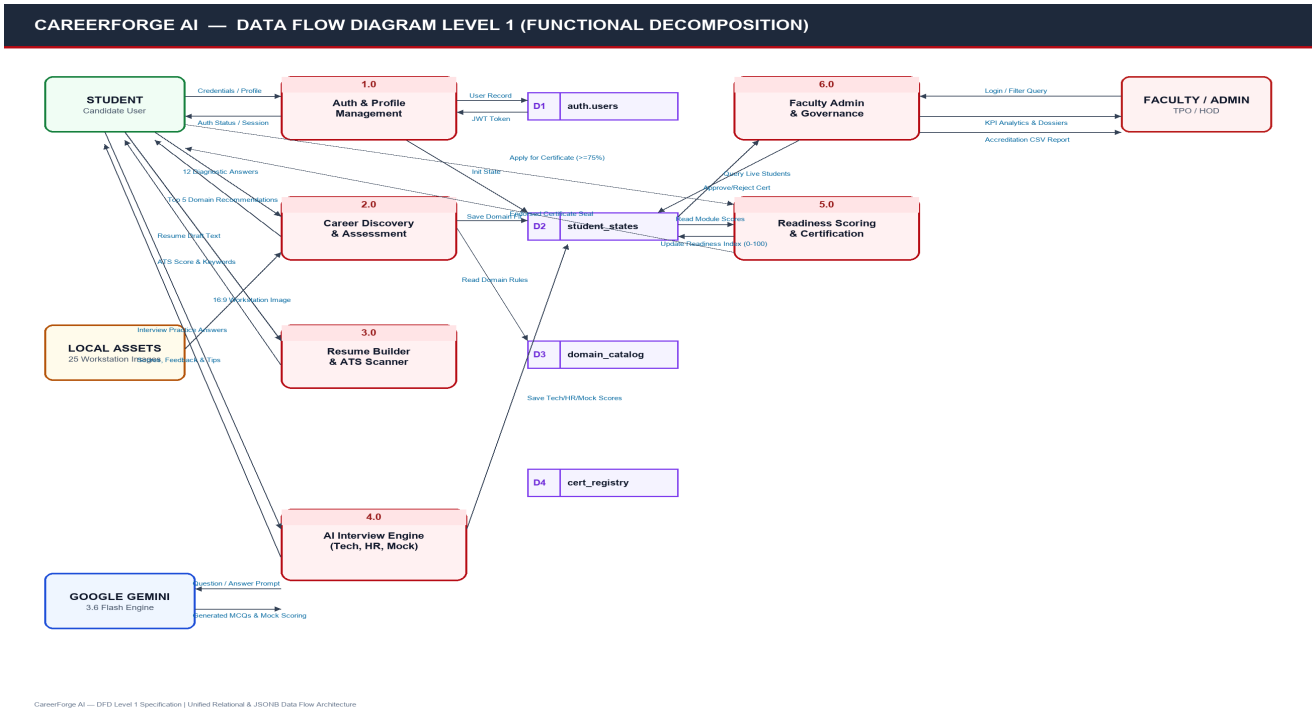


Figure 2.1: CareerForge AI — Level 1 Functional Decomposition Data Flow Diagram

Level 1 Process Decomposition Table:

Process ID	Process Name	Primary Inputs	Primary Outputs	Data Stores Interacted
1.0	**Auth & Profile Management**	Student Credentials, Name, Roll No, Dept, Year.	JWT Session Token, Initialized Student State Object.	Writes to D1: auth.users Initializes D2: student_states
2.0	**Career Discovery & Assessment**	12 Diagnostic Quiz Answers, Domain Preferences.	Top 5 Domain Recommendations, Compatibility %, 16:9 Workstation.	Reads D3: domain_catalog Updates D2: student_states
3.0	**Resume Builder & ATS Scanner**	Resume Text, Target Domain, Form Edits (3D Canvas).	ATS Score (0-100), Keyword Gap Analysis, Formatted Resume.	Reads D3: domain_catalog Updates D2: student_states
4.0	**AI Interview Engine**	Interview Mode (Tech/HR/Mock), Student Answers.	5 Live MCQs + Explanations, Mock 4-Tier Scoring, Coaching Tips.	Interacts with **Google Gemini AI** Updates D2: student_states
5.0	**Readiness & Certification**	Module Scores, Reflection Status, Certificate Application.	Readiness Index (0-100), XP/Badges, Endorsed Digital Certificate.	Reads/Updates D2: student_states Writes D4: cert_registry
6.0	**Faculty Admin & Governance**	Admin Search Query, Filter Predicates, Endorsement Decision.	Live Institutional KPIs, Student Dossiers, CSV Accreditation Export.	Queries all rows in D2: student_states

3. DFD Level 2 — Detailed Sub-Process Decomposition

Level 2 drills down into the most complex sub-system of CareerForge AI: **Process 4.0 (AI Interview Engine)**. It shows how candidate inputs, client-side pre-validation, Google Gemini LLM prompts, and anti-gibberish scoring algorithms interact internally:

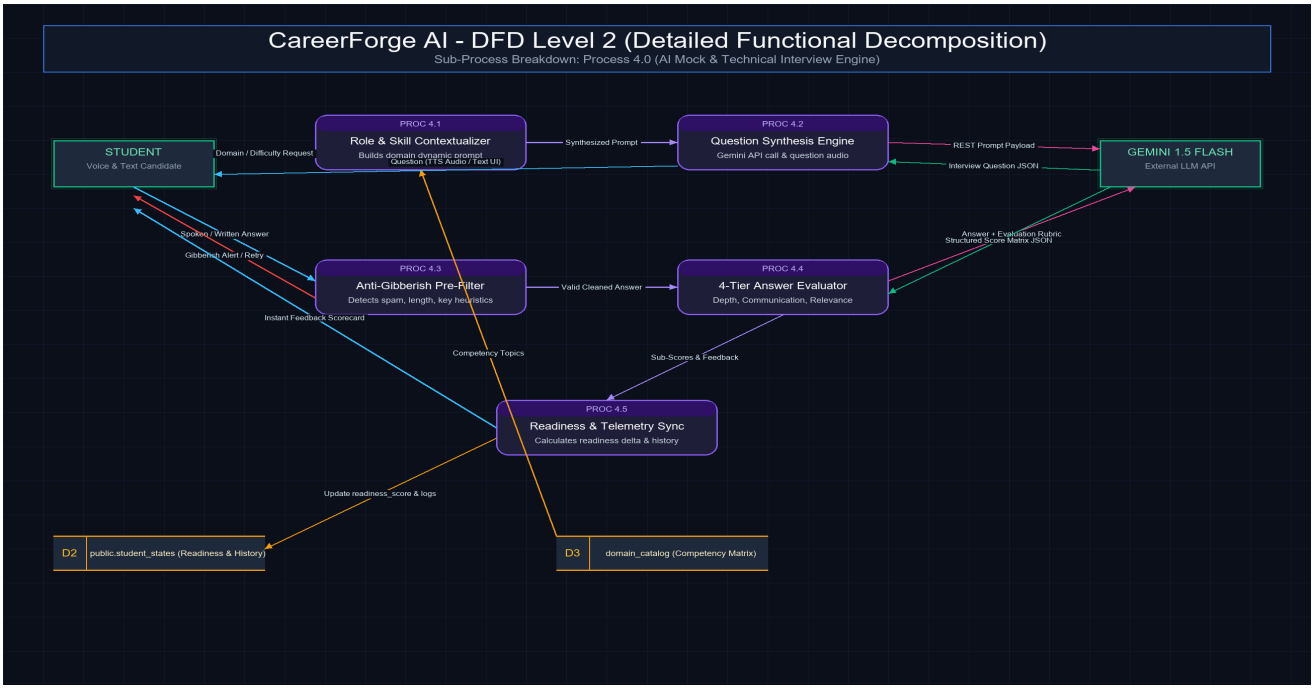


Figure 3.1: CareerForge AI — Level 2 Detailed Functional Decomposition (Process 4.0)

Sub-Process	Name & Function	Data Input Flow	Processing Logic	Data Output Flow
4.1	**Technical Question Synthesis**	Selected Domain ID (e.g. 'frontend', 'aiml', 'cloud').	Sends single compact REST prompt to Gemini 3.6 Flash acting as Principal Engineer.	5 domain MCQs with 4 options, correct index, and technical rationale.
4.2	**HR Behavioral Synthesis**	Selected Domain Name & Soft Skill Tracks.	Sends REST prompt to Gemini 3.6 Flash acting as Senior HR Director for situational prompts.	5 behavioral MCQs with recruiter reasoning and STAR interview tips.
4.3	**Anti-Gibberish Pre-Filter**	Student Written Paragraph Answer from Mock Session.	Executes isGibberishText() checking vowel ratio (<12%) and consecutive consonants.	If gibberish $\to$ flags instant 0% penalty. If valid $\to$ routes to Process 4.4.
4.4	**LLM Mock Evaluation**	Validated Candidate Written Answer & Interview Prompt.	Gemini evaluates depth and generates strict JSON with 4 distinct sub-scores.	JSON payload: *Relevance*, *Communication*, *Technical*, *Structure*, & tips.
4.5	**Feedback & Score Consolidation**	Raw Scores from 4.1, 4.2, or 4.4.	Calculates final session percentage, updates attempt count, records highest score.	Updates S.tech, S.hr, or S.mock; persists to D2: student_states.

4. Data Stores Specification (The D1-D4 Registry)

Store ID	Data Store Name	Physical Location	Schema / Structure	Access Permissions & Role

D1	**auth.users**	Supabase PostgreSQL (Cloud)	Relational table: id (UUID), email, encrypted_password, timestamps.	Managed by Supabase Auth engine. Protected by encrypted bcrypt work factors.
D2	**student_states**	Supabase PostgreSQL (Cloud)	Relational table: id (UUID PK), student_email (INDEX), state_data (JSONB).	Protected by Row Level Security (RLS). Stores complete student progress.
D3	**domain_catalog**	Local Memory / Codebase	Static array: 25 domain definitions, skills, tools, and fallback question banks.	Read-only reference store accessed during assessment and ATS scanning.
D4	**cert_registry**	Sub-document in D2	Object: { status, hash, endorsedBy, endorsedDate }.	Admin write-authorized. Unlocks verified digital placement certificate.

5. Top 10 DFD Viva Questions & Spoken Answers

Q1: What is the purpose of a Level 0 Data Flow Diagram?

Spoken Answer: A Level 0 DFD (also called a Context Level Diagram) defines the entire system boundary. It represents the whole software as a single process (0.0) and shows all external entities that supply input data or receive output responses.

Q2: What is the difference between DFD Level 0, Level 1, and Level 2?

Spoken Answer: Level 0 treats the entire software as one single process. Level 1 decomposes that process into major functional modules (like Auth, Career Discovery, AI Interviews, and Admin). Level 2 drills into a specific complex module—such as decomposing the AI Interview Engine into pre-filtering, Gemini prompt synthesis, and scoring.

Q3: What are the external entities in CareerForge AI's DFD?

Spoken Answer: There are four external entities: 1. Student (the candidate user), 2. Faculty / Admin (placement officer), 3. Google Gemini 3.6 Flash (AI engine), and 4. Supabase Cloud (PostgreSQL database and authentication).

Q4: Why are Google Gemini and Supabase treated as external entities in the DFD?

Spoken Answer: Because they exist outside the boundary of our client application code. Gemini is an external AI API service provided by Google, and Supabase is an external cloud database service that our application interfaces with via HTTPS and WebSockets.

Q5: What are the main data stores in your Level 1 DFD?

Spoken Answer: We have four data stores: D1 is `auth.users` for credentials, D2 is `public.student_states` for application progress in JSONB, D3 is `domain_catalog` for 25 career track definitions, and D4 is `cert_registry` for placement endorsement records.

Q6: How does data flow between the Student and the Admin in your DFD?

Spoken Answer: Students never communicate directly with the admin. When a student completes an interview or applies for certification, data flows from the Student to Process 5.0 and is stored in Data Store D2 (`student_states`). Process 6.0 (Admin) reads from D2 to populate the live cohort table and student dossiers.

Q7: How is the anti-gibberish filter represented in the DFD?

Spoken Answer: In our Level 2 DFD under Process 4.0, Sub-process 4.3 acts as a pre-filter. It inspects written answers for vowel ratios; if the text is random keyboard mash, it routes directly to a 0% penalty score without calling Gemini API.

Q8: What DFD notation standard are you using?

Spoken Answer: We follow the Gane and Sarson / Yourdon & Coad standard, using rounded rectangles for processes, solid rectangles for external entities, open-ended rectangles for data stores, and labeled directional arrows for data flows.

Q9: What is the balancing rule in DFDs, and did you maintain it?

Spoken Answer: The balancing rule states that all input and output data flows going into and out of a parent process in Level 0 must match the collective inputs and outputs in Level 1. We strictly preserved this balance across our Level 0 and Level 1 specifications.

Q10: Why does the 16:9 Workstation Image flow from Local Assets instead of an API?

Spoken Answer: In our Level 1 DFD, 16:9 images flow directly from local static storage (`img/`) rather than an external API. This design decision guarantees 0ms instant loading, eliminates API credit consumption, and ensures 100% offline reliability.